# F70LA – Life Insurance Mathematics A, 2025/26
# Chapter 6 – Premiums for Life Insurance Policies

**Read: Dickson, Hardy & Waters, Chapter 6**

## 6.1 Introduction

In this chapter, different ways of calculating premiums are considered. First, let us look at what the premiums need to pay for.

An insurance policy is a financial contract between the insurance company and the policyholder. The insurance company agrees to pay specific *benefits*, which are contingent on (i.e. dependent on) specified events happening. For example, that the policyholder dies in the next 20 years, or survives 15 years.

Additionally, the insurance company incurs *expenses* by selling policies. It may also wish to make a *profit*, if possible.

Meanwhile, the policyholder agrees to pay *premiums* to the insurance company to secure the benefits. The premiums should also cover the expenses and, ideally from the insurer's perspective, any profits.

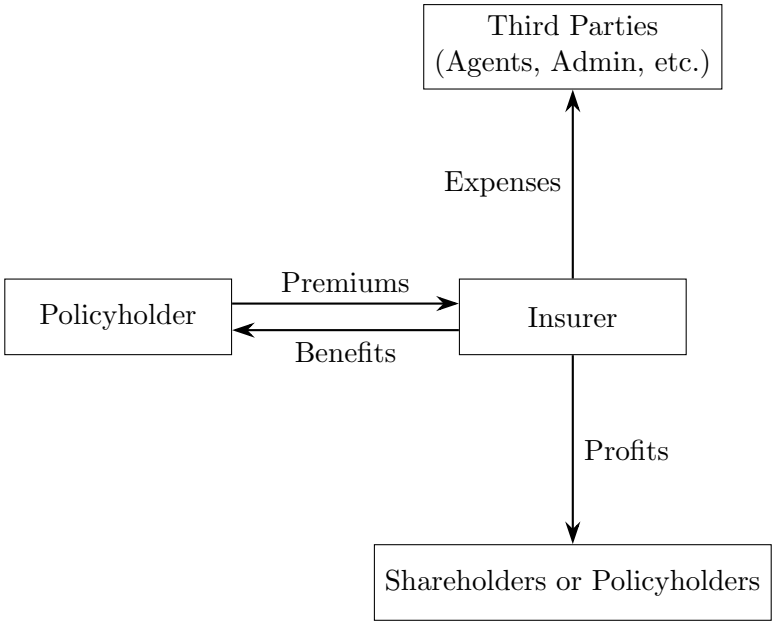



Figure 1: Cashflows due to the sale of a policy.

**Expenses** typically refer to the costs incurred by the insurer to operate the business. These expenses go to various parties, as shown in the next diagram.

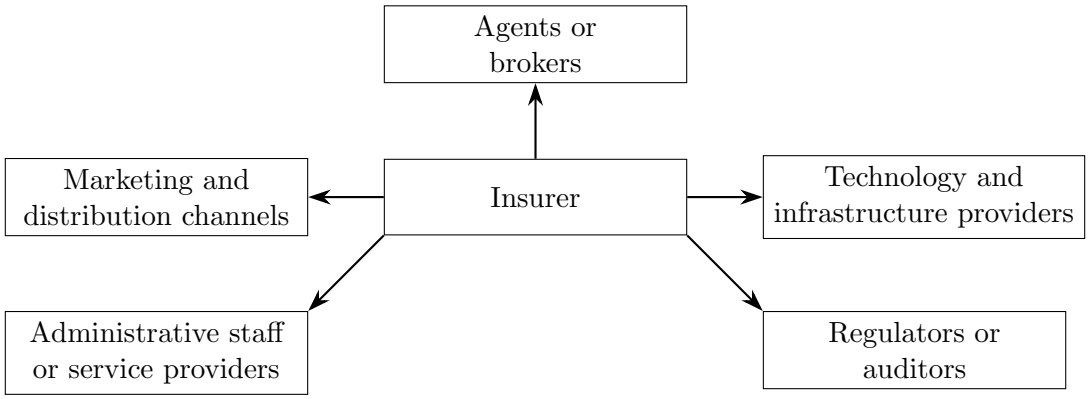



Figure 2: Some expense cashflows in a insurance company.

Some of the expenses are internal to the company (e.g. its marketing department) and others are external (e.g. an advertising agency paid to run an advertising campaign).

Who receives any **profits** depends on the type of insurance company.

- Mutual life insurance companies: A mutual insurer is owned by its policyholders. Any profits made are often returned to policyholders through dividends or lower premiums. For example, in the UK, Royal London and the National Farmers Union (NFU) Mutual Insurance Society are mutual life insurance companies.
- Stock life insurance companies: A stock insurer is owned by its shareholders. Profits are distributed to these investors. Examples in the UK include Scottish Widows, Legal & General, Zurich Insurance Group, Vitality and Aviva.

If the policyholder stops paying the premiums then the contract is broken and the policyholder is not eligible for all the benefits detailed in the insurance policy documentation.

## 6.2 Premiums

**Definition 6.1** *Premiums can either be:*

- *A **single premium**, i.e. a single payment by the policyholder. This is typical for a life annuity bought at retirement with the policyholder's retirement savings.*
- ***Regular premiums**, which are premiums paid at regular time intervals by the policyholder, such as annually, monthly, weekly, and so on.*

A feature of life insurance contracts is that premiums are payable in advance, with the first premium paid as soon as the policy is purchased.

**You should always assume in questions that premiums are payable in advance, unless explicitly told otherwise.**

It is important to charge an appropriate premium level to the customer. If the company charges too low a premium, there are a number of possible consequences:

- Underfunded liabilities – the insurer may not collect enough to cover future benefit payments.
- Regulatory issues – may lead to penalties or restrictions if reserves are insufficient.
- Unsustainable business – low premiums can erode profitability and threaten solvency.
- Cross-subsidization – losses may be offset by overcharging other policyholders, creating inequity.

Conversely, if the company charges too high a premium, it faces:

- Loss of competitiveness – customers may choose cheaper alternatives, reducing new business.
- Adverse selection – only high-risk individuals may buy the insurance product, worsening the insurer's risk pool.
- Reputational damage – overpricing can harm trust and attract regulatory scrutiny.
- Excessive profit margins – may be seen as unfair, especially in mutual or regulated environments.

There are two main types of premiums considered within the actuarial department of an insurance company, for calculating reserves and other items.

**Definition 6.2**

- *The **net premium** is the premium that covers only the insurance benefits. It does not cover expenses and profit ('net of expenses and profit'). The customer is not charged the net premium.*
- *The **gross premium**, also called the **office premium**, is the premium that includes expenses.*

## 6.3 The future loss random variable

Insurers model losses on their products for various reasons.

- Premium calculation: By modeling losses, insurers can more accurately price policies to cover potential claims and maintain profitability.
- Reserve requirements: Models help determine the capital an insurer needs to hold to pay out future claims, particularly for extreme but plausible events.
- Risk assessment and management: Insurers use loss models to understand the distribution of potential losses, identify high-risk areas and manage their exposure to catastrophic events.
- Regulatory compliance: Regulations, such as the Solvency II framework in the EU, require insurers to hold sufficient capital for plausible losses, making loss modeling essential for compliance.
- Product development: Modeling helps insurers develop new products by understanding potential risks and the costs associated with them.

Modeling allows insurers to predict the financial impact of future events by analyzing the characteristics of insurance loss distributions, such as their positive skewness and heavy tails.

**Definition 6.3** *Define the* ***net future loss*** *random variable at the start of the contract as*

$$L_0^n := PV(\textit{Future benefits}) - PV(\textit{Future net premiums})$$

*and the* ***gross future loss*** *random variable at the start of the contract as*

$$L_0^g := PV(\textit{Future benefits}) + PV(\textit{Future expenses}) - PV(\textit{Future gross premiums}).$$

Which loss random variable to use depends on the purpose of the overall calculation. If we are discussing these loss random variables in general, we use the notation $L_0$.

**Example 6.4** *Suppose a whole life insurance policy which pays the sum insured £$S > 0$ immediately upon death is sold to $(x)$. If the net premium is a single premium £$P$, then the net future loss random variable at time 0 is*

$$L_0^n = S\, v^{T_x} - P.$$

*If the net premiums are £$\tilde{P}$ paid annually as long as the policyholder is alive, then the net future loss random variable at time 0 is*

$$L_0^n = S\, v^{T_x} - \tilde{P}\, \ddot{a}_{\overline{K_x+1}|}.$$

The loss random variables allows us to calculate the premiums. There are two ways of calculating the premium that we study:

- The equivalence principle, which calculates the premium that makes the expected loss equal to zero. This approach is based on expected values and is justified by the Law of Large Numbers; and
- The portfolio percentile premium principle, which determines the premium that gives a high probability of a gain (i.e. a negative loss). It requires the full distribution of the underlying drivers of variability, such as the mortality, investment, inflation and so on.

## 6.4 The equivalence principle

To calculate a premium using the equivalence principle, set the expected value of the loss to zero, and solve for the unknown premium. Which loss random variable to use depends on whether we want to calculate a net or gross premium. We study the two calculations in turn.

### 6.4.1 Net premiums

The net premium is the premium that covers only the insurance benefits. It does not cover expenses.

**Definition 6.5** *Define the* ***net premium*** *calculated using the equivalence principle and is the solution to*

$$\mathbb{E}(PV \text{ of future net premiums}) = \mathbb{E}(PV \text{ of future benefits}).$$

*The above equation is called the* equation of value.

**Exercise 6.6** *A whole life insurance has sum insured £100 000 payable at the end of the year of death of* (40)*. Let $P$ be the net premium payable yearly.*

*The basis for calculating net premiums is the survival model AM92 Select and an interest rate of 4% per year.*

*For each question below, write down the net future loss random variable and hence calculate $P$ using the equivalence principle, assuming that:*

*(a) Premiums are payable annually for life.*

*The net future loss random variable is*

$$L_0^n = PV(\text{Future benefits}) - PV(\text{Future net premiums}) = 100\,000\, v^{K_{[40]}+1} - P\,\ddot{a}_{\overline{K_{[40]}+1|}}.$$

*Using the equivalence principle, $\mathbb{E}(L_0^n) = 0$, i.e. $100\,000\,A_{[40]} = P\,\ddot{a}_{[40]}$. Solving for $P$,*

$$P = 100\,000\,\frac{A_{[40]}}{\ddot{a}_{[40]}} = 100\,000\,\frac{0.23041}{20.009} = 100\,000\,(0.0115153) = £1,151.53.$$

(b) *Premiums are payable monthly for life.*

*The net future loss random variable is*

$$L_0^n = PV(\text{Future benefits}) - PV(\text{Future net premiums}) = 100\,000\,v^{K_{[40]}+1} - P\,\ddot{a}^{(12)}_{\overline{K_{[40]}+1|}}.$$

*Using the equivalence principle, $\mathbb{E}(L_0^n) = 0$, i.e. $100\,000\,A_{[40]} = P\,\ddot{a}^{(12)}_{[40]}$. Using Woolhouse's formula,*

$$\ddot{a}^{(12)}_{[40]} = \ddot{a}_{[40]} - \frac{11}{24} = 20.009 - \frac{11}{24} = 19.55067.$$

*Solving the equation of value for $P$,*

$$P = 100\,000\,\frac{A_{[40]}}{\ddot{a}^{(12)}_{[40]}} = 100\,000\,\frac{0.23041}{19.55067} = 100\,000\,(0.0117851) = £1,178.53.$$

(c) *Premiums are payable monthly for at most 20 years.*

*The net future loss random variable is*

$$L_0^n = PV(\text{Future benefits}) - PV(\text{Future net premiums}) = 100\,000\,v^{K_{[40]}+1} - P\,\ddot{a}^{(12)}_{\overline{\min\{K_{[40]}+1,20\}|}}.$$

*Using the equivalence principle, $\mathbb{E}(L_0^n) = 0$, i.e. $100\,000\,A_{[40]} = P\,\ddot{a}^{(12)}_{[40]:\overline{20}|}$. As*

$$v^{20}\,{}_{20}p_{[40]} = (1.04)^{-20}\,\frac{\ell_{60}}{\ell_{[40]}} = (1.04)^{-20}\,\frac{9287.2164}{9854.3036} = 0.430123,$$

*then using Woolhouse's formula,*

$$\begin{aligned}
\ddot{a}^{(12)}_{[40]:\overline{20}|} = \ddot{a}_{[40]:\overline{20}|} - \frac{11}{24}\left(1 - v^{20}\,{}_{20}p_{[40]}\right) &= \left(\ddot{a}_{[40]} - v^{20}\,{}_{20}p_{[40]}\,\ddot{a}_{60}\right) - \frac{11}{24}\left(1 - v^{20}\,{}_{20}p_{[40]}\right) \\
&= (20.009 - (0.430123)\,(14.134)) - \frac{11}{24}\,(1 - 0.430123) \\
&= 13.92964 - 0.26119 \\
&= 13.66845.
\end{aligned}$$

*Solving the equation of value for $P$,*

$$P = 100\,000\,\frac{A_{[40]}}{\ddot{a}^{(12)}_{[40]:\overline{20}|}} = 100\,000\,\frac{0.23041}{13.66845} = 100\,000\,(0.0168571) = £1,685.71.$$

*(d) Which is the lowest premium and why? Which is the highest premium and why?*

- *The lowest premium is the premium payable annually for life.*
- *£1 p.a. of annual payments in advance have a higher PV than £1 p.a. of monthly payments (i.e. £$\frac{1}{12}$ per month) in advance...*
- *..So the annual premium payable annually has to be lower than the annual premiums payable monthly, in order for them to have the same PV.*
- *The highest premium is the monthly premium payable for at most 20 years.*
- *It is higher than the monthly premium payable for life as it is payable for a shorter time period (the policyholder is age only 40 years so will live for more than 20 years with a high probability).*
- *It must be higher than the annual premium payable annually for life, using the same two arguments above.*

*(e) Would the premiums be higher if ultimate mortality was used, i.e. AM92 Ultimate, instead of select mortality? Explain your answer.*

- *The premiums are higher using the corresponding ultimate mortality.*
- *Policyholders are more likely to die under the ultimate mortality distribution,...*
- *...so are more likely to pay the premiums for a shorter time.*
- *This lowers the EPV of their premium payments.*
- *The premiums have to be larger to compensate for the lower EPV of each unit premium.*
- *You can check this by re-doing one or more of the calculations using the AM92 ultimate table and the same interest rate, e.g. the premium payable monthly for at most 20 years is about £1,687.14 per annum under AM92 ultimate, slightly higher than £1,685.71 in part (c).*

### 6.4.2 Gross premiums

The gross premium is the premium that covers the insurance benefits and expenses.

**Definition 6.7** *Define the* ***gross premium*** *calculated using the equivalence principle and is the solution to*

$$\mathbb{E}(PV \text{ of future gross premiums}) = \mathbb{E}(PV \text{ of future benefits}) + \mathbb{E}(PV \text{ of future expenses}).$$

*The above equation is also called the* equation of value.

*Profit can be included on the right-hand side of the equation of value. However, unless a question specifies otherwise, you should assume that* $\mathbb{E}(PV \text{ of future profit}) = 0$.

Before doing any gross premium calculations, we discuss expenses.

## 6.5 Types of expenses

There are different types of expenses and they can be incurred at different times. As a result, they have to be treated differently in the equation of value. The main types are:

- **Initial expenses.** Incurred by the insurer when the policy is issued. We assume in our calculations that they are incurred at the same time that the first premium is paid.
- Initial expenses include:
  - Commission payment to the agent who sold the policy to the policyholder, e.g. to a Financial Adviser.
  - Underwriting expenses, which may vary with the amount of the death benefit.
  - Cost of the setting up the policy on the insurer's computer systems.
- **Renewal expenses.** Incurred when a premium is paid and, in the case of an annuity, when an annuity payment is made.
- Examples of renewal expenses are:
  - Staff time and investment expenses caused by processing a renewal and/or annuity payments.
  - Fixed costs of the insurer, such as salaries, rent for the premises.
  - Annual statements to the policyholders.

- **Termination expenses.** Incurred when a policy expires. They are usually only a small amount.

We can put all of these into categories which are useful when applying mathematics. The categories are:

- Per policy expenses, e.g. Policy fees, settlement costs, issue costs.
- Percentage of (gross) premium expenses, e.g. Commission.
- Percentage of benefit expenses, expressed as a rate such as, e.g, £1,000 of insurance benefit. Underwriting costs would be expressed this way.

## 6.6 Allowing for profit

All the mathematics above assumed that $\mathbb{E}(\text{PV of future profit}) = 0$. However, insurance companies do make profits and this is allowed for in different ways.

We could:

- Increase the premium charged by 10%, say, over and above the premium calculated using theory;
- Use a lower rate of interest in a spreadsheet calculation of how much interest the premiums earn. This would give a higher premium; or
- Use higher than expected mortality when calculating the benefits payable for insurances, or use a lower than expected mortality when calculating the benefits payable for pensions.

There are all sorts of adjustments you could make. It is important to make sure that you know what you are adjusting so that you have a good sense of how you expect profit to emerge.

**Exercise 6.8** *Suppose a whole life policy with sum insured £100,000 is to be issued to a life aged 45. Death benefits are paid at the end of the year of death.*

*The expenses on the policy are assumed to be:*

- *75% of the 1st year's premium,*
- *5% of each year's premium thereafter.*

*Other expenses are, per £10,000 of sum insured:*

- *£10 at the beginning of the first year,*
- *£2 at the beginning of each subsequent year, and*
- *Cost of settlement of a claim is £5.*

*The gross premium basis is AM92 ultimate mortality and interest of 4% per year. Determine the annual gross premium £P payable annually for life.*

- It is important to construct a clear timeline, to help you answer the question.
- Remember, the start of the year of policyholder death is at time $K_{45}$. The benefit is paid at time $K_{45}+1$.
- The per policy expenses are expressed per £10,000 sum insured, but the policyholder's sum insured is ten times that amount. So the per policy expenses must be multiplied by 10.

| | 45 | 46 | 47 | $\cdots$ | $45+K_{45}$ | $45+K_{45}+1$ | Age |
|---|---|---|---|---|---|---|---|
| | 0 | 1 | 2 | $\cdots$ | $K_{45}$ | $K_{45}+1$ | Time |
| Premiums (£) | $-P$ | $-P$ | $-P$ | $\cdots$ | $-P$ | | |
| Benefit (£) | | | | | | $100,000$ | |
| Expenses (£) | | | | | | $+50$ | |
| | $+0.75P$ | $+0.05P$ | $+0.05P$ | $\cdots$ | $+0.05P$ | | |
| | $+100$ | $+20$ | $+20$ | $\cdots$ | $+20$ | | |

Value the premiums, benefits and expenses separately to begin with.

- EPV of expenses is

$$50\,A_{45} + 0.75P + 0.05P\,{}_{1|}\ddot{a}_{45} + 100 + 20\,{}_{1|}\ddot{a}_{45}.$$

  We can avoid calculating ${}_{1|}\ddot{a}_{45}$ - this will save us time.

**Handy hint 6.9** *Terms of the form* ${}_{1|}\ddot{a}_x$ *come up often in premium calculations. However, we can side-step their direct calculation, and instead look up only* $\ddot{a}_x$*. It is much easier, faster and less prone to error to look up* $\ddot{a}_x$ *than to have to calculate* ${}_{1|}\ddot{a}_x = v\,p_x\,\ddot{a}_{x+1}$*.*

*For example, to avoid calculating* ${}_{1|}\ddot{a}_{45}$*, consider the premium expenses in the exercise, which have an EPV of* $0.75P + 0.05P\,{}_{1|}\ddot{a}_{45}$*. The premium expense cashflows cane be written in two equivalent ways.*

| | *45* | *46* | *47* | $\cdots$ | *Age* |
|---|---|---|---|---|---|
| | *0* | *1* | *2* | $\cdots$ | *Time* |
| *Premium expenses (£)* | $+0.05P$ | $+0.05P$ | $+0.05P$ | $\cdots$ | |
| | $+0.7P$ | | | | |
| *Premium expenses (£)* | $+0.75P$ | $+0.05P$ | $+0.05P$ | $\cdots$ | |

*Motivated by the figure, write*

$$0.75P + 0.05P\,{}_{1|}\ddot{a}_{45} = (0.75P - 0.05P) + 0.05P\,\ddot{a}_{45} = 0.7P + 0.05P\,\ddot{a}_{45}.$$

*Similarly for the per policy expenses, which have EPV* $100 + 20\,{}_{1|}\ddot{a}_{45}$*, illustrate them in two equivalent ways as shown*

| | *45* | *46* | *47* | $\cdots$ | *Age* |
|---|---|---|---|---|---|
| | *0* | *1* | *2* | $\cdots$ | *Time* |
| *Per policy expenses (£)* | $+20$ | $+20$ | $+20$ | $\cdots$ | |
| | $+80$ | | | | |
| *Per policy expenses (£)* | $+100$ | $+20$ | $+20$ | $\cdots$ | |

*Motivated by the figure, write*

$$100 + 20\,{}_{1|}\ddot{a}_{45} = (100 - 20) + 20\,\ddot{a}_{45} = 80 + 20\,\ddot{a}_{45}.$$

*You will see this done repeatedly in premium calculations without any explanation. Spend some time to understand what to do and use it too!*

*It is a straightforward application of*

$$ {}_{1|}\ddot{a}_x = v\, p_x\, {}_{1|}\ddot{a}_{x+1} = 1 + v\, p_x\, {}_{1|}\ddot{a}_{x+1} - 1 = \ddot{a}_x - 1. $$

Now return to the EPV calculations.

- EPV of expenses is

  $$ 50\, A_{45} + 0.75P + 0.05P\, {}_{1|}\ddot{a}_{45} + 100 + 20\, {}_{1|}\ddot{a}_{45} = 50\, A_{45} + 0.7P + 0.05P\, \ddot{a}_{45} + 80 + 20\, \ddot{a}_{45}. $$

- EPV of premiums is $P\, \ddot{a}_{45}$.
- EPV of benefits is $100,000\, A_{45}$.

The equation of value for the annual gross premium $£P$ is

$$ \mathbb{E}(\text{PV of future gross premiums}) = \mathbb{E}(\text{PV of future benefits}) + \mathbb{E}(\text{PV of future expenses}). $$

i.e.

$$ P\, \ddot{a}_{45} = 100,000\, A_{45} + 50\, A_{45} + 0.7P + 0.05P\, \ddot{a}_{45} + 80 + 20\, \ddot{a}_{45}. $$

Solving for $P$,

$$ (1 - 0.05)\ P\, \ddot{a}_{45} - 0.7P = 100,050\, A_{45} + 80 + 20\, \ddot{a}_{45} \quad \Rightarrow \quad P = \frac{100,050\, A_{45} + 80 + 20\, \ddot{a}_{45}}{(1 - 0.05)\ \ddot{a}_{45} - 0.7}. $$

Looking up the AM92 Ultimate tables at an interest rate of 4% p.a.,

$$ P = \frac{100,050\,(0.27605) + 80 + 20\,(18.823)}{(1 - 0.05)\ (18.823) - 0.7} = \frac{28,075.2625}{17.18185} = £1,634.01. $$

**Exercise 6.10** *Calculate the single gross premium for a deferred annuity of £80,000 per year payable monthly in advance, issued to a select life now aged 50 with the first annuity payment on the life's 65th birthday.*

*Assume:*

- *Initial expenses of £1,000.*
- *Renewal expenses on each anniversary of the issue date, provided the policyholder is alive.*
- *The renewal expenses will be £20 on the first policy anniversary date, and will increase with inflation from that date at the rate of 1.923% per year.*
- *Survival model AM92 Select.*
- *Interest: 6% per year.*

Set $g = 0.0193$.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 50 | 51 | 52 | $\cdots$ | 64 | 65 | $65\frac{1}{12}$ | $65\frac{2}{12}$ | $\cdots$ | 66 | $\cdots$ | Age |
| | 0 | 1 | 2 | $\cdots$ | 14 | 15 | $15\frac{1}{12}$ | $15\frac{2}{12}$ | $\cdots$ | 16 | $\cdots$ | Time |
| Premium (£) | $-P$ | | | | | | | | | | | |
| Benefit (£) | | | | | | $\frac{80{,}000}{12}$ | $\frac{80{,}000}{12}$ | $\frac{80{,}000}{12}$ | $\cdots$ | $\frac{80{,}000}{12}$ | $\cdots$ | |
| Expenses (£) | $1{,}000$ | 20 | $20(1+g)$ | $\cdots$ | $20(1+g)^{13}$ | $20(1+g)^{14}$ | | $\cdots$ | | $20(1+g)^{15}$ | $\cdots$ | |

- EPV of expenses is

$$\begin{aligned}
&1{,}000 + 20\left(v\,p_{[50]} + (1+g)\,v^2\,{}_2p_{[50]} + (1+g)^2\,v^3\,{}_3p_{[50]} + \cdots\right)\\
=&1{,}000 - \frac{20}{1+g} + 20\left(\frac{1}{1+g} + v\,p_{[50]} + (1+g)\,v^2\,{}_2p_{[50]} + (1+g)^2\,v^3\,{}_3p_{[50]} + \cdots\right)\\
=&1{,}000 - \frac{20}{1+g} + \frac{20}{1+g}\left(1 + (1+g)\,v\,p_{[50]} + (1+g)^2\,v^2\,{}_2p_{[50]} + (1+g)^3\,v^3\,{}_3p_{[50]} + \cdots\right)\\
=&1{,}000 - \frac{20}{1+g} + \frac{20}{1+g}\,\ddot{a}_{[50]},
\end{aligned}$$

  in which $\ddot{a}_{[50]}$ is calculated at the annual effective interest rate of $\frac{1.06}{1+g} - 1 = \frac{1.06}{1.01923} - 1 = 0.0400$. Using AM92 Select, the EPV of expenses is

$$1{,}000 - \frac{20}{1.01923} + \frac{20}{1.01923}\,(17.454) = £1{,}322.87.$$

- EPV of the single premium is $P$.

- EPV of benefits is $80{,}000\,{}_{15|}\ddot{a}^{(12)}_{50}$. Using Woolhouse's formula and AM92 Select mortality and an annual effective interest rate of 6%,

$$\begin{aligned}
80{,}000\,{}_{15|}\ddot{a}^{(12)}_{50} = 80{,}000\,v^{15}\,{}_{15}p_{[50]}\,\ddot{a}^{(12)}_{65} =& 80{,}000\,(1.06)^{-15}\,\frac{\ell_{65}}{\ell_{[50]}}\left(\ddot{a}_{65} - \frac{11}{24}\right)\\
=& 80{,}000\,(1.06)^{-15}\,\frac{8821.2612}{9706.0977}\left(12.276 - \frac{11}{24}\right)\\
=& 80{,}000\,(0.379226)\;(11.81767)\\
=& £358{,}525.27.
\end{aligned}$$

The equation of value for the annual gross premium £$P$ is

$$\mathbb{E}(\text{PV of future gross premiums}) = \mathbb{E}(\text{PV of future benefits}) + \mathbb{E}(\text{PV of future expenses}).$$

i.e.

$$P = 358{,}525.27 + 1{,}322.87 = £359{,}848.14.$$

## 6.7 Traditional insurance contracts

Life insurance policies can be categorised into two types:

- Without profit (also called non-participating). So far, we have seen only without profit contracts in this course. These are policies in which the sum assured is predetermined when the contract is issued. For example, it is a constant £100,000 or it will increase by £10,000 each year with absolute certainty.

- Unit-linked policy. The sum insured in a unit-linked policy is linked directly to the investment return achieved on the invested premiums. The insurance company has no discretion to adjust the sum insured in a different way.

- With profits (also called non-participating). In with profits contracts, the profits earned on invested premiums are shared with the policyholders. However, the profits do not have to be all shared immediately with the policyholders, unlike with a unit-linked policy. Some of the profits can be withheld for the future.

  In North America, the profits are used to pay cash dividends or reduce the premiums. In the UK, the traditional approach is use the profits to increase the sum insured. We study the UK traditional approach in this course.

  When the with profit contract is issued, it is not known how much the profits will be during the lifetime of a policy. So none of these additional benefits are guaranteed at the policy inception date.

| Always assume a contract is without profit, unless explicitly told otherwise. |
|---|

## 6.8 With profits (Participating) Contracts

With profits contracts are those where the sum assured increases throughout the policy's term, through bonuses called *reversionary bonuses* and *terminal bonuses*.

Reversionary bonuses are awarded during the term of the policy. Once a **reversionary bonus is awarded, it is guaranteed**. It is usual that reversionary bonuses can never be negative, i.e. the sum assured will not decrease due to a bonus.

Terminal bonuses are awarded when the policy matures.

**Remark 6.11** *From the 1970s to 1990s in the UK, with profit endowment insurance policies were very common, taken out at the same time as a mortgage and with the same term. The policyholder would pay only the interest on the mortgage, and not make any capital repayments. They also paid premiums to the endowment insurance policy, with the expectation that the sum assured at the maturity date would cover the capital outstanding on the mortgage plus it would generate an additional cash surplus (through a terminal bonus which you learn more about next). The underlying premiums were invested partially in the stock market. However, since the* mortgage endowment mis-selling scandal*, which you can*

*easily read more about using the internet, with profit contracts have fallen out of favour in the UK. However, with profit contracts are still popular in many countries.*

### 6.8.1 Bonuses on with profits contracts

When a with profit contract is issued, it has a **basic** or **original** sum assured attached to it, e.g. £100,000. The expectation is that the sum assured will increase over the term of the contract. However, during the first policy year, the sum assured remains £100,000. So if the policyholder dies during the first year, £100,000 is the amount of death benefit.

#### Reversionary bonuses

At the start of each policy year - perhaps including the first policy year, perhaps not - a reversionary bonus is declared. The largest driver of the level of bonus is generally the investment returns. – the better they have been, the higher the reversionary bonus. Other factors include the claims experience, e.g. how many policyholders died and hence a claim was made, versus how many were expected to die, and the expenses experience - actual versus expected expenses, and so on.

**Definition 6.12** *reversionary bonuses can be expressed in three different ways, called* bonus mechanisms*:*

- ***Simple reversionary bonus**, when the **original** sum assured only is increased by a fixed percentage each year.*

  *E.g. suppose the basic sum insured is £$S = 100,000$. At the start of each policy year, except the first, a simple bonus rate of $b = 4\%$ is declared. The table shows when the sum insured is increased and to what amount.*

| *Time* $n$ | 0 | 1 | 2 | $\cdots$ | 25 |
|---|---|---|---|---|---|
| *Sum insured (£)* $= S \times (1 + bn)$ | $100,000$ | $100,000$ $\times(1 + 1 \times 0.04)$ $= 104,000$ | $100,000$ $\times(1 + 2 \times 0.04)$ $= 108,000$ | $\cdots$ $\cdots$ $\cdots$ | $100,000$ $\times(1 + 25 \times 0.04)$ $= 200,000$ |

  *The policyholder who dies during the first year will be paid £100,000. The policyholder who dies during the second year will be paid £104,000, and so on. The policyholder who dies in the last year, after time 24 and just before time 25, will be paid £196,000. Only the policyholders who survive the entire 25 years will get £200,000.*

- ***Compound reversionary bonus***, *when the sum assured and any previously declared bonuses are increased by a fixed percentage each year.*

  *E.g. suppose the basic sum insured is* $£S = 100,000$. *At the start of each policy year – apart from the first – a compound bonus rate of* $\tilde{b} = 2.8114\%$ *is declared.*

| *Time* $n$ | 0 | 1 | 2 | $\cdots$ | 25 |
|---|---|---|---|---|---|
| *Sum insured (£)* | | $100,000$ | $100,000$ | $\cdots$ | $100,000$ |
| $= S \times (1+\tilde{b})^n$ | | $\times(1.028114)$ | $\times(1.028114)^2$ | $\cdots$ | $\times(1.028114)^{25}$ |
| | $100,000$ | $= 102,811$ | $= 105,702$ | $\cdots$ | $= 200,000$ |

- ***Super compound reversionary bonus*** *- This is as for compound reversionary bonus except that the basic sum assured is increased at a different rate to that of any previously declared bonuses.*

  *E.g. suppose the basic sum insured is* $£S = 100,000$. *At the start of each policy year, except the first, a super-compound bonus rate of* $b_1 = 2\%$ *is declared on the basic sum assured and a super-compound bonus rate of* $b_2 = 5.35\%$ *is declared on the past bonuses.*

| *Time* $n$ | 0 | 1 | 2 | $\cdots$ | 25 |
|---|---|---|---|---|---|
| *Sum insured (£)* | | $100,000$ | $100,000$ | $\cdots$ | $100,000$ |
| $= S(1+b_1 s_{\overline{n}\vert}^{b_2})$ | | $\times(1+0.02)$ | $\times(1+0.02)$ | $\cdots$ | $\times(1+0.02 s_{\overline{25}\vert}^{5.35\%})$ |
| | $100,000$ | $= 100,000$ | $+2,000 \times 0.0535$ | $\cdots$ | $= 200,192$ |
| | | $+2,000$ | $= 100,000$ | $\cdots$ | |
| | | $= 102,000$ | $+4,107$ | $\cdots$ | |
| | | | $= 104,107$ | $\cdots$ | |

*Reversionary bonuses* vest *at the start of each policy year. Whether this includes the first year or not depends on the insurer's practice and the wording of the policy.* Vest *means that the declared bonus becomes guaranteed and irrevocably part of the policy benefits.*

*Future expected reversionary bonuses are valued when calculating premiums. This is because policyholders' have a reasonable expectation that such bonuses are declared regularly.*

**Remark 6.13** *Due to the judicious choice of the bonus rates, the sum insured in all three methods end up at the same value in the numerical examples above. However, they grow at different rates before the maturity date.*

*The growth of the sum insured has implications for the risk and return of the with profits contracts. For example, the sum insured is larger at earlier durations for the simple rate method. As the sum insured is guaranteed to be paid, this means the insurance company would take less investment risk on the invested premiums of a simple rate contract. In turn, this may mean a lower returns overall for the policyholder (although, we constructed our examples above all to have the same return at the end of the contract).*

### Terminal bonus

A **terminal bonus** is declared at the policy maturity date as a lump sum amount, either at the maturity date or upon the death of the policyholder. The terminal bonus due to death is awarded only if investment returns have been sufficient to the date of death. THe earlier the death during the term, the less likely that investment returns will have been sufficient.

The policyholder (or their nominated beneficiary) receives the basic sum assured, all the declared reversionary bonuses and the terminal bonus.

The terminal bonus is not directly allowed for when setting the premium. This is because they are non-guaranteed and discretionary, and effectively less discretionary than reversionary bonuses. Policyholders' reasonable expectation is considered to be that a terminal bonus reflects long-term investment performance, which is volatile.

**Exercise 6.14** *The basic sum insured for a 20-year with profits endowment insurance is £100 000. The sum insured is payable at the end of the year of death or at maturity. Reversionary bonuses vest at the start of each policy year, including the first.*

*Premiums of amount £P per year are payable monthly in advance throughout the term of the policy.*

*Calculate P for a life aged 40 assuming:*

*(a) The bonuses are simple reversionary bonuses at the rate of 2% per year.*

*(b) The bonuses are compound reversionary bonuses at the rate of 1.923% per year.*

*Basis: Survival model: AM92 Select. Interest: 6% per year.*
*Expenses: 50% of the first premium and 10% of all remaining premiums.*

(a) The bonuses are simple reversionary bonuses at the rate of 2% per year, which means £2,000 is added to the basic sum assured at the start of each policy year, including the first.

- First,
$$v^{20}\,\frac{\ell_{60}}{\ell_{[40]}} = (1.06)^{-20}\,\frac{9,287.2164}{9,854.3036} = 0.29386.$$
- EPV of premiums are
$$P\,\ddot{a}^{(12)}_{[40]:\overline{20}|} = P\left(\ddot{a}_{[40]:\overline{20}|} - \frac{11}{24}\left(1 - v^{20}\,\frac{\ell_{60}}{\ell_{[40]}}\right)\right) = P\left(12.000 - \frac{11}{24}\,(1 - 0.29386)\right) = 11.676P.$$
- EPV of benefits is $100,000\,A_{[40]:\overline{20}|} + 2,000\,(IA)[40]:\overline{20}|$. Applying Exercise 4.57,
$$\begin{aligned}(IA)_{[40]:\overline{20}|} = (IA)^{1}_{[40]:\overline{20}|} + v^{20}\,{}_{20}p_{[40]} &= (IA)_{[40]} - v^{20}\,{}_{20}p_{[40]}\;((IA)_{60} + 20A_{60}) + v^{20}\,{}_{20}p_{[40]}\\ &= (IA)_{[40]} - v^{20}\,{}_{20}p_{[40]}\;((IA)_{60} + 20A_{60} - 1)\\ &= 3.85489 - 0.29386\;(5.46572 + 20(0.32692) - 1)\\ &= 0.62121.\end{aligned}$$

The EPV of benefits is thus

$$100,000\, A_{[40]:\overline{20}|} + 2,000\, (IA)_{[40]:\overline{20}|} = 100,000\,(0.32076) + 2,000\ (0.62121) = 33,318.41.$$

– EPV of expenses is

$$0.5\frac{P}{12} + \left(0.1P\, \ddot{a}^{(12)}_{[40]:\overline{20}|} - 0.1\frac{P}{12}\right) = 0.4\frac{P}{12} + 11.676\,(0.1P) = 1.201\, P.$$

The equation of value is

$$\text{EPV of premiums} = \text{EPV of benefits} + \text{EPV of expenses},$$

i.e.

$$11.676P = 33,318.41 + 1.201\, P \quad \Rightarrow \quad P = £3,180.76.$$

(b) The bonuses are compound reversionary bonuses at the rate of 1.923% per year vesting at the start of each policy year. The EPV of the premium and expenses are unchanged from part (a); it is only the EPV of the benefits which needs re-worked.

In this case, the EPV of benefits is

$$\begin{aligned}
&100,000 \left[ \left( \frac{1.01923}{1.06}\, q_{[40]} + \left(\frac{1.01923}{1.06}\right)^2 {}_{1|}q_{[40]} + \cdots + \left(\frac{1.01923}{1.06}\right)^{20} {}_{19|}q_{[40]} \right) \right. \\
&\qquad \left. + \left(\frac{1.01923}{1.06}\right)^{20} {}_{20}p_{[40]} \right] \\
=&100,000 \left[ \left( \frac{1}{1.04}\, q_{[40]} + \left(\frac{1}{1.04}\right)^2 {}_{1|}q_{[40]} + \cdots + \left(\frac{1}{1.04}\right)^{20} {}_{19|}q_{[40]} \right) + \left(\frac{1}{1.04}\right)^{20} {}_{20}p_{[40]} \right] \\
=&100,000\, A_{[40]:\overline{20}|} \quad \text{valued at } 4\% \text{ p.a.} \\
=&100,000(0.46423) \\
=&46,423.
\end{aligned}$$

The equation of value is

$$\text{EPV of premiums} = \text{EPV of benefits} + \text{EPV of expenses},$$

i.e.

$$11.676P = 46,423 + 1.201\, P \quad \Rightarrow \quad P = £4,431.79.$$

## 6.9 The Portfolio Percentile Premium Principle

The **portfolio percentile premium principle** is an alternative to the equivalence principle. The portfolio percentile premium principle is to choose the premium so that there is a specified probability that the insurer's profit is positive.

Suppose the insurer has a portfolio of $N > 0$ policies, written on $N$ different lives each with an i.i.d. future lifetime. Denote by $L_{0,i}$ the future loss random variable of the $i$th policy, for $i = 1, 2, \ldots, N$. Denote by $L$ the total future loss in the portfolio of policies, i.e.

$$L := \sum_{k=1}^{N} L_{0,i}.$$

The insurer chooses a number $\alpha \in (0, 1)$ and wishes to choose the premium paid by each policyholder so that

$$\boxed{\mathbb{P}[\text{Positive profit on the total portfolio}] = \mathbb{P}[\text{Negative loss}] = \mathbb{P}[L < 0] = \alpha.}$$

How to calculate the probability of a profit? From the Central Limit Theorem, as $N$ gets larger, the total loss $L$ is approximately normally distributed with mean $\mathbb{E}(L)$ and variance $\mathbb{V}ar(L)$. This is because it is the sum of $N$ i.i.d. random variables. We assume

$$\boxed{L = \sum_{k=1}^{N} L_{0,i} \sim N\left(\mathbb{E}(L), \mathbb{V}ar(L)\right).}$$

Thus, letting $\Phi$ denote the cumulative distribution function of a standard normal random variable,

$$\alpha = \mathbb{P}[L < 0] = \mathbb{P}\left[\underbrace{\frac{L - \mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}}}_{\text{Standard normal r.v.}} < \frac{0 - \mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}}\right] = \Phi\left(-\frac{\mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}}\right).$$

Denoting the inverse of $\Phi$ by $\Phi^{-1}$, and applying it both sides of the last approximation gives

$$\Phi^{-1}(\alpha) = -\frac{\mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}}.$$

The last task is for the insurer to choose the premiums paid by the $N$ policyholders – which affect the mean and variance of $L$ – so that the last expression holds.

**Exercise 6.15** *Show that*

$$\Phi^{-1}(\alpha) = -\sqrt{N}\,\frac{\mathbb{E}(L_{0,1})}{\sqrt{\mathbb{V}ar(L_{0,1})}}.$$

First note that, as $L_{0,1}, L_{0,2}, \ldots, L_{0,N}$ are i.i.d. random variables,

$$\mathbb{E}(L) = \sum_{k=1}^{N} \mathbb{E}(L_{0,i}) = \sum_{k=1}^{N} \mathbb{E}(L_{0,i}) = N\,\mathbb{E}(L_{0,1}) \quad \text{and} \quad \mathbb{V}ar(L) = \sum_{k=1}^{N} \mathbb{V}ar(L_{0,i}) = N\,\mathbb{V}ar(L_{0,1}).$$

Then

$$\Phi^{-1}(\alpha) \approx -\frac{\mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}} = -\frac{N\,\mathbb{E}(L_{0,1})}{\sqrt{N\,\mathbb{V}ar(L_{0,1})}} = -\sqrt{N}\,\frac{\mathbb{E}(L_{0,1})}{\sqrt{\mathbb{V}ar(L_{0,1})}}.$$

**Remark 6.16** *As $N$ gets larger, the premium calculated becomes smaller. As $N \to \infty$, the premium should tend to the equivalence principle premium.*

*The equivalence principle assumes that idiosyncratic mortality risk has been perfectly diversified, shown by its use of the survival and mortality probabilities in its 'per policy' premium calculations. Doing the premium calculations across a portfolio of $N$ i.i.d. policies, achieved by multiplying all terms by $N$, shows that the number of deaths assumed in the equivalence principle-based calculation is, e.g. $Nq_x$ in the first year. This is an assumption of 'perfect pooling' of idiosyncratic mortality risk.*

*In contrast, the portfolio percentile premium principle allows for idiosyncratic mortality risk, i.e. variability in the number of deaths against the expected number. What if $N$ is small, so the number of deaths has a reasonable chance of being something other than, e.g. $Nq_x$ in the first year?*

*As the number of policies becomes larger, idiosyncratic mortality risk is increasingly reduced. This means that the actual number of deaths will tend to the expected number of deaths (along each sample path as well as in the observed number of deaths, if our basis reflects reality). So for a premium calculated using the portfolio percentile premium principle and considered as a function of $N$, it should tend to the premium calculated using the equivalence principle as $N \to \infty$.*

**Exercise 6.17** *A life office issues 1,000 identical whole life insurances with sum insured £100 000, payable at the end of the year of death, to lives aged 60. Future lifetimes are assumed to be independent random variables.*

*A level annual premium is payable throughout life. The office uses the following basis for calculating premiums:*

*Survival model: AM92 Select*
*Interest: 4% per year*
*Expenses: Nil.*

*(a) Calculate the level annual premium $\bar{P}$ using the equivalence principle.*

$$\bar{P}\,\ddot{a}_{[60]} = 100\,000\,A_{[60]} \quad \Rightarrow \quad \bar{P} = 100\,000\,\frac{A_{[60]}}{\ddot{a}_{[60]}} = 100\,000\,\frac{0.45510}{14.167} = £3,212.40.$$

(b) *Calculate the level annual premium $P$ such that there is a 90% probability that the present value of the profit on this portfolio of policies is positive.*

*Let $L$ denote the total loss on the portfolio and set $N := 1,000$, the number of policies. We want to calculate the premium $P$ so that $\mathbb{P}[L < 0] = 0.9$, i.e. a 90% chance of a profit.*

*Remember the key result, and derive what you need from it:*

$$L = \sum_{k=1}^{1,000} L_{0,i} \sim N\left(\mathbb{E}(L), \mathbb{V}ar(L)\right).$$

*For the $i$th portfolio, $L_{0,i} = 100,000\, v^{K_{[60]}+1} - P\, \ddot{a}_{\overline{K_{[60]}+1|}}$, in which, to reduce notation, we omit an indicator to show that $K_{[60]}$ is the curtate future lifetime r.v. of the $i$th policyholder.*

*The mean of the total loss is $\mathbb{E}(L) = \sum_{k=1}^{N} \mathbb{E}(L_{0,i}) = \sum_{k=1}^{N} \mathbb{E}(L_{0,i}) = N\,\mathbb{E}(L_{0,1})$, with*

$$\mathbb{E}(L_{0,1}) = 100,000\, A_{[60]} - P\, \ddot{a}_{[60]} = 100,000\,(0.45510) - P\,(14.167) = 45,510 - 14.167P.$$

*The variance of the total loss is $\mathbb{V}ar(L) = \sum_{k=1}^{N} \mathbb{V}ar(L_{0,i}) = N\,\mathbb{V}ar(L_{0,1})$, where*

$$\begin{aligned}
\mathbb{V}ar(L_{0,1}) = \mathbb{V}ar\left(100,000\, v^{K_{[60]}+1} - P\, \ddot{a}_{\overline{K_{[60]}+1|}}\right) &= \mathbb{V}ar\left(100,000\, v^{K_{[60]}+1} - P\,\frac{1 - v^{K_{[60]}+1}}{d}\right) \\
&= \mathbb{V}ar\left(\left(100,000 + \frac{P}{1-v}\right) v^{K_{[60]}+1} - \frac{P}{d}\right) \\
&= \left(100,000 + \frac{P}{1-v}\right)^2 \mathbb{V}ar\left(v^{K_{[60]}+1}\right) \\
&= \left(100,000 + \frac{P}{1-v}\right)^2 \left({}^2A_{[60]} - \left(A_{[60]}\right)^2\right) \\
&= \left(100,000 + \frac{P}{1-v}\right)^2 (0.168386)^2.
\end{aligned}$$

*Then*

$$\begin{aligned}
0.9 = \mathbb{P}[L < 0] = \mathbb{P}\left[\frac{L - \mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}} < \frac{0 - \mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}}\right] &= \Phi\left(-\frac{\mathbb{E}(L)}{\sqrt{\mathbb{V}ar(L)}}\right) \\
\Rightarrow \Phi^{-1}(0.9) = -\frac{N(45,510 - 14.167P)}{\sqrt{N}\,\left(100,000 + \frac{P}{1-v}\right)(0.168386)}&
\end{aligned}$$

*Rearrange the last expression to solve for $P$, i.e.*

$$\left(100,000 + \frac{P}{1-v}\right)(0.168386)\,\Phi^{-1}(0.9) = -\sqrt{N}\,(45,510 - 14.167P)$$

*i.e.*

$$\left(14.167\sqrt{N} - \frac{(0.168386)\,\Phi^{-1}(0.9)}{1-v}\right) P = (100,000)\,(0.168386)\,\Phi^{-1}(0.9) + 45,510\,\sqrt{N}.$$

*Given that* $\Phi^{-1}(0.9) = 1.2816$ *and* $N = 1,000$*, the premium is*

$$P = \frac{21,580.35 + 45,510\,\sqrt{1,000}}{14.167\sqrt{1,000} - 5.6109} = £3,301.92.$$

**Remark 6.18** *In the last exercise, how does the premium calculated using the portfolio percentile premium principle change as $N$ varies? Consider this premium as a function of the number of policies $N$, i.e.*

$$P(N) = \frac{21,580.35 + 45,510\,\sqrt{N}}{14.167\sqrt{N} - 5.6109}.$$

| $N$ | $P(N)$ |
|---|---|
| 1 | 7,841 |
| 10 | 4,223 |
| 100 | 3,503 |
| 1,000 | 3,302 |
| 10,000 | 3,240 |
| 100,000 | 3,221 |

*The premiums become smaller as $N$ increases. At $N = 1,000$, the premium of £3,302 becomes similar to the equivalence principle premium of £3,212, albeit still higher which reflects the less-than-perfect mortality pooling, compared to a smaller number of policies.*

**Remark 6.19** *We have assumed that investment returns are known and constant. Thus randomness comes only from deaths not occurring exactly in line with the survival model. In practice, random investment returns add another layer of risk. They would be incorporated into the loss random variable by replacing the constant interest rate $i$ by a random variable $i_t$ at each future time $t$ (so $\{i_t\}_t$ is a stochastic process).*